

Maximum & Minimum

Name: _____

Pre-Calculus | Guided notes — we fill these in together

Vocabulary

Max / min: the _____ of a parabola (the vertex); opens up gives a _____, opens down gives a _____.

Increasing / decreasing: read left to right — the graph goes _____ (increasing) or _____ (decreasing).

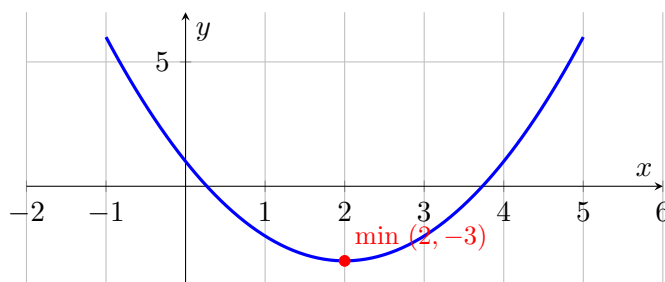
We do together. For $y = (x - 2)^2 - 3$, give the vertex, state max or min, and the intervals.

Step 1. Vertex form $a(x - h)^2 + k$ gives vertex _____.

Step 2. Leading coefficient is +1 (opens up), so the vertex is a _____.

Step 3. It decreases for _____ and increases for _____.

See it:



opens up $\Rightarrow$ minimum at $(2, -3)$.

You Try. Solve each. Show every step, then name the answer.

a) Give the vertex and max/min of $y = -x^2 + 4$.

b) Give the intervals of $y = x^2$.

Summary (fill in): Opens _____ $\Rightarrow$ minimum; opens _____ $\Rightarrow$ maximum; the graph turns at the _____.